Domain and Range

My Definition

Facts & Characteristics

Domain

Examples

Non-examples

My Definition

Facts & Characteristics

Range

Examples

Non-examples

Contracts

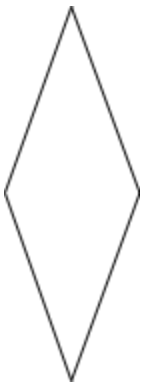
Contracts tell us how to use a function. For example: `ellipse :: (Number, Number, String, String) -> Image` tells us that the name of the function is `ellipse`, it takes four inputs (two Numbers and two Strings), and it evaluates to an `Image`. From the contract, we know `ellipse(50, 100, "solid", "teal")` will evaluate to an `Image`.

Name		Domain		Range
`# triangle`	`::`	`Number, String, String`	`->`	`Image`
`triangle(80, "solid", "darkgreen")`				
`# star`	`::`		`->`	
`# circle`	`::`		`->`	
`# square`	`::`		`->`	
`# rectangle`	`::`		`->`	
`# rhombus`	`::`		`->`	
`# ellipse`	`::`		`->`	
`# text`	`::`		`->`	
`# regular-polygon`	`::`		`->`	
`#right-triangle`	`::`		`->`	

Using Contracts (continued)

Use the contracts to write expressions to generate images similar to those pictured.

`rhombus :: Number, Number, String, String -> Image`

	
Write an expression for a square (rotated) using `rhombus`!	
What variable changes with the first Number?	
What variable changes with the second Number?	

Triangle Contracts (SAS & ASA)

Type each expression (left) below into the editor and match it to the image it creates (right).

Expression			Image
`triangle-sas(120, 45, 70, "solid", "black")`	1	A	
`triangle-sas(120, 90, 70, "solid", "black")`	2	B	
`triangle-sas(120, 135, 70, "solid", "black")`	3	C	
`triangle-sas(70, 135, 120, "solid", "black")`	4	D	

Think about how you would describe each of the arguments that `triangle-sas` takes in to someone who'd never used the function before and annotate the contract below using descriptive variable names.

```
triangle-sas :: (
    ____________________________________________ :: Number,
    ____________________________________________ :: Number,
    ____________________________________________ :: Number,
    ____________________________________________ :: String,
    ____________________________________________ :: String
) -> Image
```

Add examples of each of the triangle functions we've explored to your contracts page.

If you have time, experiment with the `triangle-asa` function.

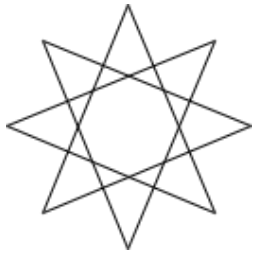
```
triangle-asa :: (
    left-angle :: Number,
    left-side :: Number,
    bottom-angle :: Number,
    style :: String
    color :: String
) -> Image
```

Star Polygon

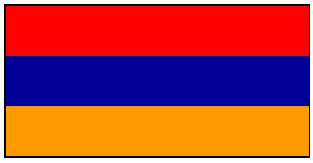
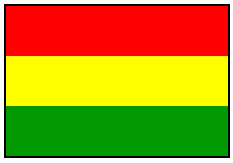
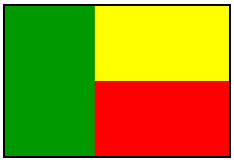
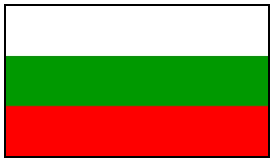
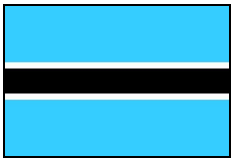
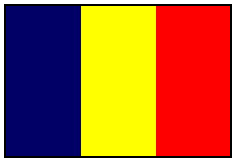
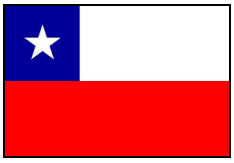
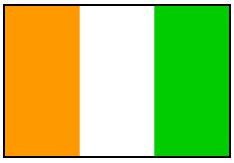
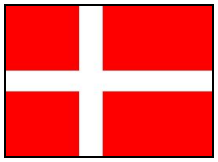
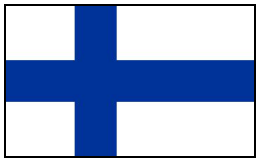
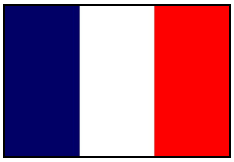
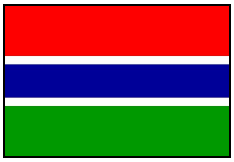
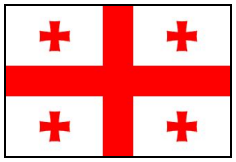
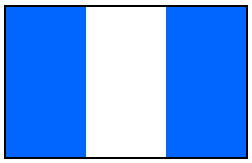
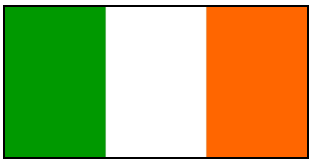
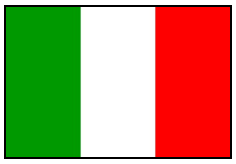
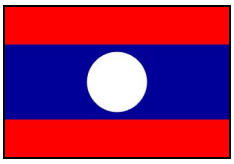
```
star-polygon :: (

    side-length :: Number,
    points-on-polygon :: Number,
    polygon-points-to-skip-between-star-points :: Number,
    shading-style  :: String,
    color :: String

) -> Image
```

Using the detailed contract above, write expressions to create each image below.

Then make two more star polygons of your choosing. Sketch them and write expressions to generate them.

	_____
	_____
	_____
	_____

These flags can all be created using the basic shape functions in addition to overlay, beside, and above.

 Armenia (1:2)	Belgium (13:15)	 Bolivia (15:22)	 Benin (2:3)
 Bulgaria (3:5)	 Botswana (2:3)	Burkina Faso (2:3)	Cameroon (2:3)
 Chad (2:3)	 Chile (2:3)	Costa Rica (3:5)	 Cote d'Ivoire (2:3)
 Denmark (28:37)	Estonia (7:11)	 Finland (11:18)	 France (2:3)
Gabon (3:4)	 The Gambia (2:3)	 Georgia (2:3)	Germany (3:5)
Ghana (2:3)	Greece (2:3)	 Guatemala (5:8)	Guinea-Bissau (1:2)
Hungary (1:2)	 Ireland (1:2)	 Italy (2:3)	 Laos (2:3)

Writing Examples from Purpose Statements 2

We've provided contracts and purpose statements to describe two different functions. Write examples for each of those functions.

Contract and Purpose Statement

Every contract has three parts…

half-image:: *String* -> *Image*

function name *Domain* *Range*

Consumes an image, and produces that image scaled to half its size.

what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is**

function name *input(s)*

_____

what the function produces

_____ (_____) **is**

function name *input(s)*

_____

what the function produces

end

Contract and Purpose Statement

Every contract has three parts…

product-squared:: *Number, Number* -> *Number*

function name *Domain* *Range*

Consumes two numbers and squares their product

what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

_____ (_____) **is** _____

function name *input(s)* *what the function produces*

end

Rocket Height Challenges

1) Can you make the rocket fly faster?

2) Can you make the rocket fly slower?

3) Can you make the rocket sink down instead of fly up?

4) Can you make the rocket accelerate over time, so that it moves faster the longer it flies?

5) Can you make the rocket blast off and then land again?

6) Can you make the rocket blast off, reach a maximum height of exactly 1000 meters, and then land?

7) Can you make the rocket blast off, reach a maximum height of exactly 1000 meters, and then land after exactly 100 seconds?

8) Can you make the rocket fly to the edge of the the universe?

Blank Design Recipe

Directions:

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions:

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Restaurants)

Directions: Use the Design Recipe to write a function `split-tab` that takes in a cost and the number of people sharing the bill and splits the cost equally.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: Use the Design Recipe to write a function `tip-calculator` that takes in the cost of a meal and returns the 15% tip for that meal.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Direct Variation)

Directions: Use the Design Recipe to write a function `wage`, that takes in a number of hours worked and returns the amount a worker will get paid if their rate is $10.25/hr.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: On average, people burn about 11 calories/minute riding a bike. Use the Design Recipe to write a function `calories-burned` that takes in the number of minutes you bike and returns the number of calories burned. .

Contract and Purpose Statement

Every contract has three parts…

_____ :: *Number* -> *Number*
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Slope/Intercept 1)

Directions: For his birthday, James' family decided to open a savings account for him. He started with $50 and committed to adding $10 a week from his afterschool job teaching basketball to kindergartners. Use the Design Recipe to write a function `savings` that takes in the number of weeks since his birthday and calculates how much money he has saved.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: Use the Design Recipe to write a function `moving` that takes in the days and number of miles driven and returns the cost of renting a truck. The truck is $45 per day and each driven mile is 15¢.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Negative Slope/Intercept)

Directions: An Olympic pool holds 660,000 gallons of water. A fire hose can spray about 250 gallons per minute. Use the Design Recipe to write a function `pool` that takes in the number of minutes that have passed and calculates how much water is still needed to fill it.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: The community arts fund awards a $1500 grant each month to support a new mural. They started with $50000 in their account. Use the Design Recipe to write a function `funds-available` that takes in the number of months and calculates how much money they have left.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Geometry - Rectangles)

Directions: Use the Design Recipe to write a function `lawn-area` that takes in the length and width of a rectangular lawn and returns its area.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: Use the Design Recipe to write a function `rect-perimeter` that takes in the length and width of a rectangle and returns the perimeter of that rectangle.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Geometry - Rectangular Prisms)

Directions: Use the Design Recipe to write a function `rectprism-vol` that takes in the length, width, and height of a rectangular prism and returns the Volume of a rectangular prism.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: Use the Design Recipe to write a function `rect-prism-sa` that takes in the width, length and height of a rectangular prism and calculates its surface area (the sum of the areas of each of its six faces)

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is**
function name *input(s)*

_____
what the function produces

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Geometry - Circles)

Directions: Use the Design Recipe to write a function `circle-area-dec` that takes in a radius and uses the decimal approximation of pi (3.14) to return the area of the circle.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: Use the Design Recipe to write a function `circumference` that takes in a radius and uses the decimal approximation of pi (3.14) to return the circumference of the circle.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Geometry - Cylinders)

Directions: Use the Design Recipe to write a function `circle-area` that takes in a radius and uses the fraction approximation of pi ($\frac{22}{7}$) to return the area of the circle.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: Use the Design Recipe to write a function `cylinder` that takes in a cylinder's radius and height and calculates its volume, making use of the function *circle-area*.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Breaking Even)

Directions: The Swamp in the City Festival is ordering t-shirts. The production cost is $75 to set up the silk screen and $9 per shirt. Use the Design Recipe to write a function `min-shirt-price` that takes in the number of shirts to be ordered, *n*, and returns the minimum amount the festival should charge for the shirts in order to break even. (Assume that they will sell all of the shirts.)

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

The Design Recipe (Marquee & Cubing)

Directions: Use the Design Recipe to write a function `marquee` that takes in a message and returns that message in large gold letters.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Directions: Use the Design Recipe to write a function `num-cube` that takes in a number and returns the cube of that number.

Contract and Purpose Statement

Every contract has three parts…

_____ :: _____ -> _____
function name *Domain* *Range*

_____
what does the function do?

Examples

Write some examples, then circle and label what changes…

examples:

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

_____ (_____) **is** _____
function name *input(s)* *what the function produces*

end

Definition

Write the definition, giving variable names to all your input values…

fun _____ (_____):
function name *variable(s)*

_____
what the function does with those variable(s)

end

Vocabulary Practice

Below is a new structure definition:

```
data MediaType:
  | book(
      title :: String,
      author :: String,
      pubyear :: Number)
end

# an example book:
book1 = book("1984", "Orwell", 1949)
```

Fill in the blanks below with the vocabulary term that applies to each name. Here are the terms to choose from:

contract	example
header	field
data type	instance
constructor	data block
name	purpose

`author` is a _____

`book` is a _____

`MediaType` is a _____

`book1` is a _____

`title` is a _____

`data` … **`end`** is a _____